The Relationship Between State Education Finance and Student Achievement:

An Application of the Generalized Linear Model

Eileen (Ziyan) Li, Siyuan Zheng

University of Pennsylvania, School of Social Policy & Practice

MSSP 8970: Applied Linear Modeling

Instructor: Dr. Momin Malik

Policy Research Report 4

December 17, 2025

Introduction

Education finance policy in the United States is shaped by a long-standing empirical debate over the magnitude and structure of the relationship between public education funding and student achievement. On one hand, several studies find positive effects of increased school spending. Differences in research design, levels of aggregation, and model specification partly explain these conflicting findings. Studies place spending in a different light and suggest that it alone may be insufficient to generate meaningful gains in academic performance (Hanushek, 2003; Jackson et al., 2016). At the same time, states vary substantially in their reliance on federal education revenue, which is often targeted toward higher-need populations and may therefore be associated with lower observed achievement levels.

This study examines these relationships using a structured sequence of linear regression models, in which model specification, diagnostic evaluation, and robustness checks are treated as integral components of the analytic process. Based on NAEP scores from 1992 to 2015, this study addresses two primary questions:

1. Do models examining the relationship between education finance and student achievement meet the assumptions of the general linear model?
2. When diagnostic issues are addressed through model correction, how do the estimated relationships between instructional spending, federal revenue reliance, and NAEP performance change?

A secondary question builds on these goals:

3. When diagnostic issues are addressed, what relationships remain between instructional spending, federal revenue reliance, and NAEP performance?

Background

The allocation of spending in education has long pitted researchers against policymakers. Early literature included the findings of Hanushek (2003), which suggested that spending does not necessarily lead to better outcomes, with more recent quasi-experimental studies finding the meaningful effect of increased spending on education and economic outcomes to be true (Jackson et al., 2016). The allocation of spending in the context of education in the United States also tends to be quite decentralized.

Specifically, federal education funding is sometimes targeted towards states and districts with higher need levels via Title I, for example. Thus, states with larger federal funding proportions might systematically vary from other states in unobservable ways with regard to factors like poverty levels, demographics, and past investments (Baker, 2018). As a result, simple correlations between federal revenue share and student achievement are likely to reflect targeting and selection mechanisms rather than the causal effect of federal funding itself. The observed inverse relationships between federal funding and student performance might not represent the causal effect of federal funding but might be due to other factors like targeting and selection.

This study will conceptualize federal revenue share not as a policy failure but as an indicator of structural disadvantage and will also investigate whether Federal Funding mitigates or interacts with Instructional Expenses in determining achievement outcomes. By situating education spending within broader fiscal structures, this study contributes to the literature by shifting the focus from whether spending matters to how funding composition shapes the effectiveness of instructional investment.

Methods

Data

The analysis uses data from the U.S. Education Datasets: Unification Project, a publicly available, harmonized state-year dataset that integrates information on education finance, enrollment, and student achievement in the United States. The unit of analysis is the state-year.

The analytic sample includes all U.S. states with available National Assessment of Educational Progress (NAEP) data between 1992 and 2015. After restricting the sample to years with valid outcome measures and applying listwise deletion for missing values on key analytic variables, the final dataset consists of 564 state-year observations.

Variables

All variables used in this analysis were drawn directly from the *U.S. Education Datasets: Unification Project* and were either used in their original form or constructed through transparent transformations in R. Variable names referenced below correspond exactly to variables appearing in the dataset or in the analytic code.

Outcome Variable

Student academic achievement is measured using a composite National Assessment of Educational Progress (NAEP) score. This variable, labeled naep_composite in the analytic dataset, was constructed in R as the row-wise mean of four NAEP state-level average scores available in the dataset:

- **avg_math_4_score:** the state's average score for fourth graders taking the NAEP math exam.

- **avg_math_8_score:** the state's average score for eight graders taking the NAEP math exam.
- **avg_reading_4_score:** the state's average score for fourth graders taking the NAEP reading exam.
- **avg_reading_8_score:** the state's average score for eighth graders taking the NAEP reading exam.

Key Independent Variables

Instructional Spending per Pupil (log-transformed) (log_instr_pp)

Instructional spending per pupil captures the level of state investment in classroom instruction. This variable was constructed in two steps using the following dataset variables:

- **instruction_expenditure:** total state instructional spending
- **grades_all:** the count of all students in the state. Comparable to ENROLL in the financial data (which is the U.S. Census Bureau's estimate for students in the state).

The logarithmic transformation was applied to reduce right skewness in the distribution of per-pupil spending and to better approximate linearity between instructional investment and student achievement.

Federal Revenue Share (fed_share)

Federal reliance is measured using the proportion of total education revenue derived from federal sources. This variable, labeled fed_share in the analytic dataset, was constructed as:

fed_share = federal_revenue / total_revenue

where:

- **federal_revenue** represents total federal education revenue received by the state
- **total_revenue** represents total state education revenue from all sources

Values of fed_share range from 0 to 1, with higher values indicating greater reliance on federal funding. This variable is therefore interpreted as capturing structural fiscal reliance and underlying need rather than the causal effect of federal education policy.

Control Variables

- Year fixed effects to account for national trends in testing standards, inflation, and education policy.

Analytic Strategy

This paper uses a general linear model (GLM) framework to explore the relationship between state education finance and NAEP student achievement. We use ordinary least squares regression (OLS), which is a special case of GLM with a Gaussian error distribution and an identity link function. This statistical approach suits the level of measurement (continuous) and distribution (approximately normal) of the dependent variable (NAEP_composite).

We focus on diagnostic checking and model building before substantive interpretation. The analysis consists of four steps: specification of a baseline regression model, checking linear model assumptions, detecting and addressing influential observations, and model moderation.

Base Regression Model

The base regression model estimates the association between student achievement and education finance variables:

$$NAEP_composite\ (s,t) = \beta_0 + \beta_1 \times log_instr_pp\ (s,t) + \beta_2 \times fed_share\ (s,t) + \gamma t + \varepsilon$$

Where:

NAEP_composite (s, t): the composite NAEP score in state s and year t.

Log_instr_pp (s, t): log-transformed instructional spending per pupil.

Fed_share (s, t): the proportion of education revenue derived from federal sources.

γt: year fixed effects.

ε(s,t) is the error term.

Evaluation of Linear Model Assumptions

Model diagnostics were conducted to assess the key assumptions of the general linear model, including linearity, homoskedasticity, normality of residuals, and the presence of influential observations. After specifying the regression baseline, we check the following key linear model assumptions. First, linearity is checked using scatterplots of naep_composite showing fitted linear regression lines and LOESS smoothed lines. Then, we check homoskedasticity using residuals vs. fitted plots and plots of absolute values of residuals. We check normality of residuals using residuals histograms and Q-Q plots, and finally, we check multicollinearity using VIF scores.

In sum, diagnostics suggested moderate heteroskedasticity and a number of high leverage observations with no multicollinearity (VIF < 1).

Identification and Treatment of Influential Observations

We identify influential observations using Cook's distance, leverage values, and studentized residuals. A limited number of state-year observations, almost exclusively from the District of Columbia, influence the model parameters calculated using the regression.

Instead of excluding these observations, we re-estimate models after dropping influential observations to check for robustness. The model fits much better after exclusion, as shown by a substantial decrease in AIC values. The implicated observations are attached to the observations for DC. All subsequent modeling is based on the updated electronics sample.

Moderation Analysis

To test whether associations between instruction spending and NAEP scores depend on the amount of federal revenue states receive, we specify an interaction term with log_instr_pp and fed_share in the updated regression baseline. The rationale for the interaction is that for states with a higher share of federal revenue, the association between instruction expenditures and NAEP is higher. We will test for the need for the interaction using a nested F-test and AIC values for statistical comparison.

Results

Descriptive Statistics

Table 1 displays the descriptive statistics on the variables employed in the regression analysis. There are 564 state-year data points in the analytical sample. This data ranges over the years 1992-2015.

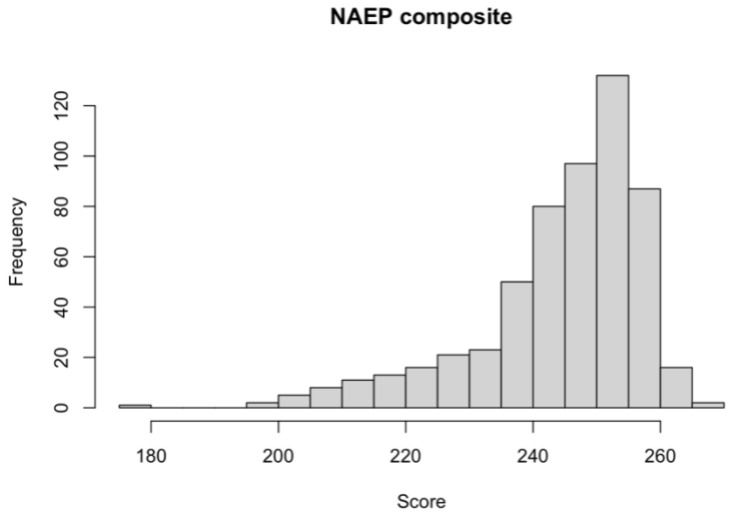
Table 1. *Descriptive Statistics*

variable	N	mean	sd	min	max
fed_share	564	0.0902849	0.0337030	0.02917083	0.2228299
log_instr_pp	564	1.5743196	0.3731209	0.59564347	2.7390315

variable	N	mean	sd	min	max
naep_composite	564	244.5585106	13.1791812	179.00000000	266.0000000

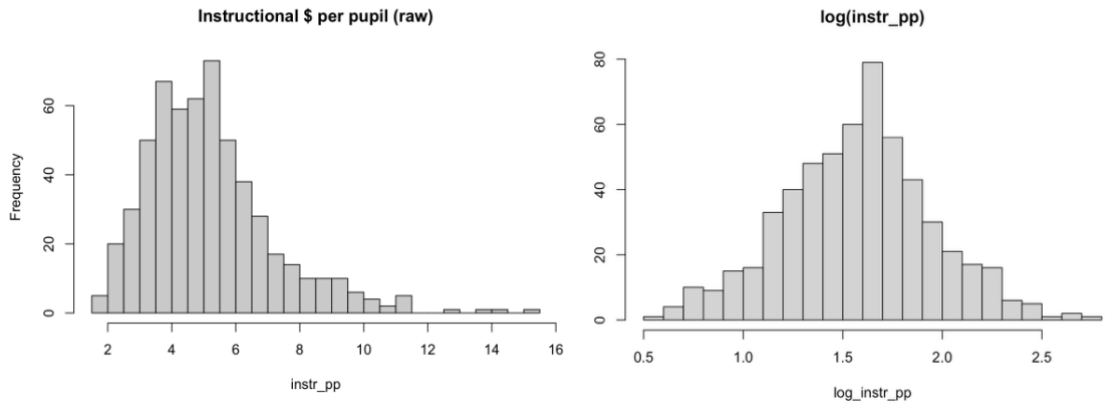
The mean NAEP composite score is 244.56 with a standard deviation of 13.18, and observed values range from 179 to 266. Visual inspection of the histogram suggests an approximately normal distribution, supporting the use of linear regression methods.

Figure 1. *NAEP composite distribution*



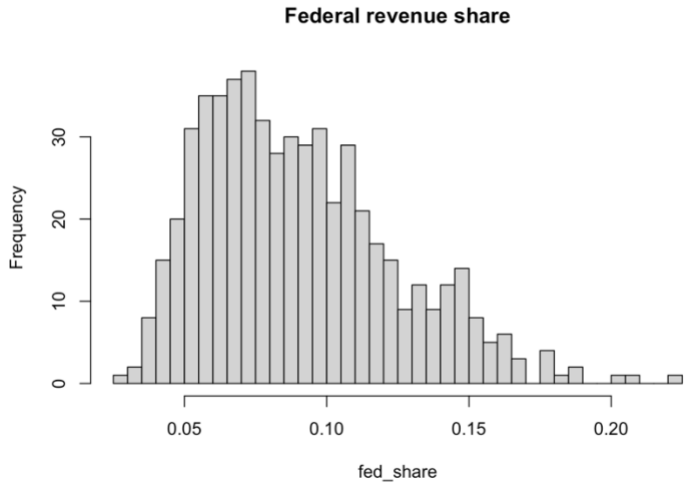
The average log-transformed instructional expenses per pupil (log_instr_pp) is 1.57 (SD = 0.37), ranging from 0.60 to 2.74. The histogram of instructional spending per pupil prior to transformation exhibits substantial right skewness, providing empirical justification for the log transformation used in subsequent regression models.

Figure 2. *Spending per pupil distribution (raw vs log)*



The mean federal share (fed_share) is 0.09, and it varies between 0.03 and 0.22, indicating a significant difference among the states concerning the use of federal funding for educational purposes. The distribution of the federal share is moderately right-skewed but restricted to valid levels.

Figure 3. *Federal Share Distribution*



In general, it appears that there is sufficient variation regarding descriptive statistics and distributions that suggest multivariate regression is possible for both education finance variables and student achievement.

Correlation Analysis

The following is a description of Table 2, showing Pearson correlation coefficients amongst key analytical variables.

Table 2. *Correlation Matrix*

	naep_composite	log_instr_pp	fed_share
naep_composite	1.000	0.648	-0.057
log_instr_pp	0.648	1.000	-0.083
fed_share	-0.057	-0.083	1.000

The composite score offered by NAEP (naep_composite) is strongly and positively correlated with log spending on instruction per pupil (log_instr_pp) ($r = 0.65$). This indicates that states with higher instructional spending per pupil tend to exhibit higher NAEP scores in

bivariate relationships. As expected, these bivariate associations do not account for time trends, unobserved heterogeneity, or potential confounding factors addressed in the multivariate models.

Similarly, federal share (fed_share) is weakly negatively correlated with NAEP achievement ($r = -0.06$). This implies that those states that rely on federal aid more tend to have a slightly low achievement score.

The correlation coefficient between log_intr_cc and fed_share is small ($r = -0.08$), suggesting there is very little relationship between this predictor variable and the predictor variable "fed_share". In general, it seems that the correlation matrix does not indicate multicollinearity in subsequent regression analysis.

Base Regression Model

Table 3 presents estimates from the baseline regression model relating NAEP composite scores to instructional spending per pupil and federal revenue share, controlling for year fixed effects. The base regression models relate student outcomes to the variables describing education finance, along with year dummies. These models are a starting point for diagnostic checking and further development.

Table 3. *Base Model Regression*

Model 1 Results

	Dependent variable:
	naep_composite
log_instr_pp	0.895
	(1.257)
fed_share	-133.042***
	(10.301)
factor(year)1994	-20.497***
	(1.391)

factor(year)1998	2.072
	(1.423)
factor(year)2000	14.489***
	(1.465)
factor(year)2002	7.239***
	(1.509)
factor(year)2003	16.401***
	(1.519)
factor(year)2005	18.324***
	(1.606)
factor(year)2007	19.114***
	(1.649)
factor(year)2009	20.801***
	(1.742)
factor(year)2011	25.438***
	(1.916)
factor(year)2013	22.468***
	(1.754)
factor(year)2015	20.425***
	(1.759)
Constant	241.922***
	(2.048)
Observations	564
R^2	0.778
Adjusted R^2	0.773
Residual Std. Error	6.283 (df = 550)
F Statistic	148.248*** (df = 13; 550)
Note:	*p<0.1; **p<0.05; ***p<0.01

In the basic model, the federal revenue share (fed_share) has a strong negative relationship with the NAEP composite score ($\beta = -133.04$, $p < .001$). Substantively, a one

percentage point increase in the share of education revenue derived from federal sources is associated with an approximately 1.33-point decrease in the NAEP composite score, holding instructional spending and year effects constant.

On the other hand, the log of instructional expenditures per pupil (log_instr_pp) has a positive but not statistically significant association with student achievement ($\beta = 0.90$, $p = .48$). This suggests that, in the baseline specification, instructional spending per pupil does not exhibit a statistically detectable association with NAEP performance once federal revenue share and time trends are accounted for.

The year fixed effects are statistically significant for most years, suggesting that there are persistent gains in NAEP scores over time compared to the base year. The model has a large amount of variation in student outcomes explained (Adjusted $R^2 = 0.77$), although this could potentially be due to the presence of influence points and violated assumptions.

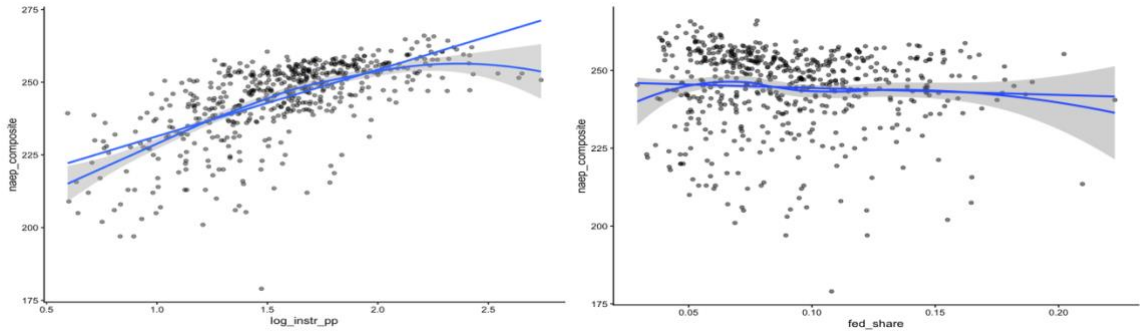
Given that regression diagnostics indicate that there are influential observations, it is important to check how robust these results are when a modified model is considered.

Evaluation of Linear Model Assumptions

After estimation of the basic regression model, diagnostics were run to check some of the key assumptions of the general linear model.

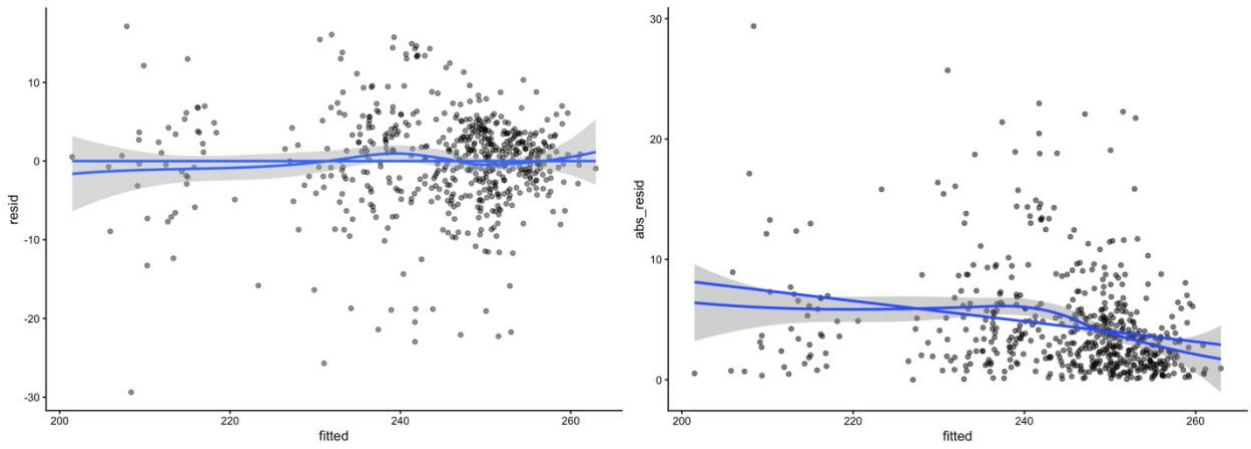
The linearity condition was checked through the use of a scatter plot between NAEP's composite score and the log of spending per student on an instructional budget, on which a linear fit and a LOESS smoother are placed. It can be seen in Figure 4 that the LOESS fit closely tracks along with the linear fit over most of the range, which confirms that a linear relationship between spending and academic performance can be correctly specified by a linear equation.

Figure 4. *Linearity Plots*



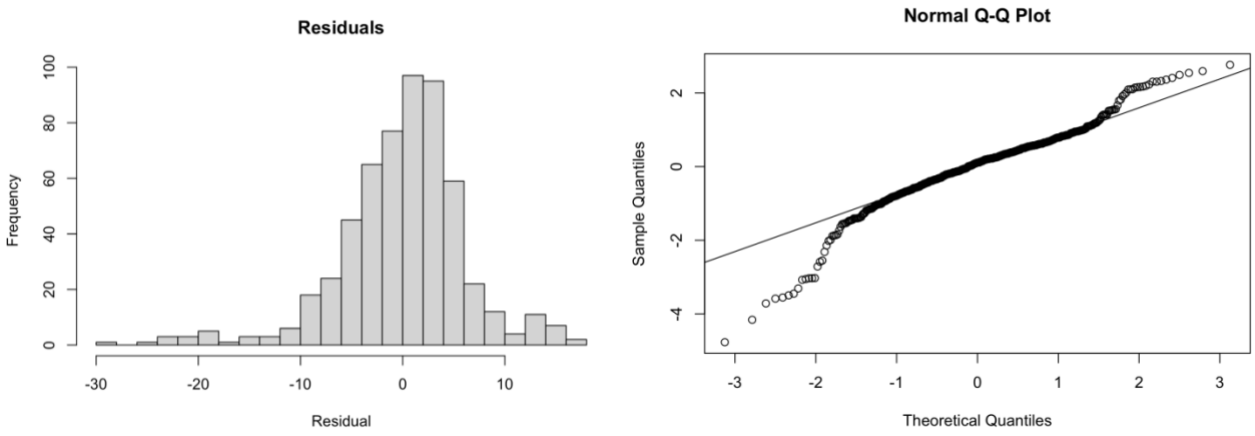
The homoskedasticity was tested using residual plots versus the fit plots and the plot of absolute residuals. The results indicate that there is the slightest indication of heteroskedasticity, where the variance in the residuals increases with the increase in the fit values. While the diagnostics do not indicate severe heteroskedasticity, they suggest mild deviations from constant variance that warrant caution in interpretation.

Figure 5. *Homoskedasticity*



The normality of residuals was checked through residual plots and Q-Q plots. The residuals are almost normally distributed, but with some slight irregularities in the upper end. The irregularities are found at points where there are some extreme values.

Figure 6. *Normality*



Finally, influential observations were assessed using Cook's distance, leverage values, and studentized residuals. As shown by the influence diagnostics, several state-year observations exert disproportionate influence on the base model estimates, most prominently repeated observations from the District of Columbia and a small number of outlying state-years such as Michigan in 1998.

Table 4. *Outliers*

state	year	cooks_d	leverage	stud_resid
DISTRICT_OF_COLUMBIA	1994	0.06506371	0.03851105	-4.866172
DISTRICT_OF_COLUMBIA	2000	0.04402218	0.07719112	-2.730269
DISTRICT_OF_COLUMBIA	1992	0.04204952	0.03290128	-4.222990
DISTRICT_OF_COLUMBIA	2003	0.03269484	0.03210302	-3.758964
DISTRICT_OF_COLUMBIA	1998	0.02559322	0.05089377	-2.598428

Overall, diagnostic test outcomes show that although the basic model satisfies linear model assumptions largely, it is necessary to check cases because of heteroskedasticity and data points with high leverage. This supports the analysis and development of robustness tests, to be followed in the next section.

Corrected Base Regression Model

To check the reliability of the regression findings, the base model was also re-estimated after eliminating state-year observations identified as influential in Cook's distance, leverage-based values, and studentized residuals. As a part of diagnostic analysis, most of these influential observations during repeated observations from the District of Columbia were observed, while a small number of additional outliers could be found from other states, such as Michigan in 1998. The corrected model should be noted that it does have the same specification as the base model for log instructional spending per pupil, federal revenue share, and year fixed effects; however, it is estimated on a smaller sample of analytic data.

The state-year observation count of 564 drops to 522 after the removal of influential observations. It improves considerably as seen in the methodology. Model fit improves substantially following the removal of influential observations, as evidenced by a sharp decline in the Akaike Information Criterion from 3689.46 in the baseline model to 3027.86 in the corrected specification, which can only be explained by a better fit to the empirical data.

Table 5. *AIC*

	df	AIC
m1	15	3689.459
m1_corr	15	3027.858

The corrected base model's conclusions are only very marginally different in a practical sense from those implied by the original specification. The negative association of federal revenue share (fed_share) with student achievement is still very significant and strong (β = -154.34, $p < .001$). In fact, the roughly 1.54-point decrease in the NAEP composite score could be linked to each additional percentage point increase in the share of education revenue coming from federal sources, considering both instructional spending and year fixed effects at the same time. The size of the association implied by the corrected model is markedly larger than that in

the base model, which suggests that the influential observations in the original sample partially blocked the influence of the unobserved variable.

Table 6. *Corrected Based Regression*

	Dependent variable:
	naep_composite
log_instr_pp	0.999
	(0.949)
fed_share	-154.336***
	(8.629)
factor(year)1994	-20.382***
	(1.012)
factor(year)1998	3.383***
	(1.030)
factor(year)2000	14.461***
	(1.054)
factor(year)2002	8.312***
	(1.110)
factor(year)2003	15.787***
	(1.098)
factor(year)2005	17.728***
	(1.168)
factor(year)2007	19.024***
	(1.210)
factor(year)2009	20.919***
	(1.298)
factor(year)2011	26.426***
	(1.457)
factor(year)2013	22.977***

	(1.302)
factor(year)2015	20.467***
	(1.299)
Constant	243.572***
	(1.588)
Observations	522
R^2	0.875
Adjusted R^2	0.872
Residual Std. Error	4.333 (df = 508)
F Statistic	273.812*** (df = 13; 508)
Note:	*p<0.1; **p<0.05; ***p<0.01

Log instructional spending per pupil retains a small positive but statistically non-significant association with NAEP performance ($\beta = 1.00$, $p = 0.293$), indicating that spending levels alone do not robustly predict achievement once influential observations are addressed. This actually points to the fact that despite the eliminated influencing observations, instructional spending per student on its own is still not able to bring an improvement in overall student achievement irrespective of the general time trends.

It is noteworthy that the fixed effects model tracked statistically significant patterns in most years and indicated NAEP performance that was slightly higher over the years compared to that in the reference year. The corrected model shows a total proportion of variance accounted for of 0.872, proving that the increased explanatory power of the corrected model is based on real improvements of the model rather than overfitting.

Thus, the corrected base model is consistent with the first regression results. But, because of the errors of the original estimation in the first model, confidence in the results has been strengthened and is no longer rather sensitive to state-year observations which were influential. Therefore, a secure basis for inspection of whether instructional spending and student

achievement are specifically moderated by federal education funding similarity among states, is for sure. The next moderation analysis involves the interplay between federal education funding similarity and state instructional spending.

Moderation Analysis: Interaction Between Instructional Spending and Federal Revenue Share

To explore whether the relationship between instructional spending per pupil and student achievement varies by states' reliance on federal education funding, an interaction term between log instructional spending per pupil × federal revenue share was added to the corrected base regression model. This specification tests whether instructional spending is more strongly associated with NAEP performance in states that receive a larger share of education revenue from federal sources.

Results from the interaction model indicate that the interaction term between log_instr_pp and fed_share is positive and statistically significant ($\beta = 41.10$, $p = .016$). This result would suggest that the relationship between instructional spending and student achievement varies systematically by the level of federal revenue reliance. More particularly, instructional spending is more positively related to NAEP composite scores in states with higher federal revenue shares.

Table 7. *Interaction Model*

	Dependent variable:
	naep_composite
log_instr_pp	-2.418
	(1.701)
fed_share	-219.688***
	(28.380)
factor(year)1994	-20.339***

	(1.008)
factor(year)1998	3.452***
	(1.026)
factor(year)2000	14.590***
	(1.050)
factor(year)2002	8.545***
	(1.109)
factor(year)2003	16.033***
	(1.097)
factor(year)2005	17.974***
	(1.167)
factor(year)2007	19.240***
	(1.208)
factor(year)2009	21.047***
	(1.293)
factor(year)2011	26.316***
	(1.450)
factor(year)2013	23.180***
	(1.298)
factor(year)2015	20.726***
	(1.297)
log_instr_pp:fed_share	41.099**
	(17.011)
Constant	248.919***
	(2.720)
Observations	522
R^2	0.877
Adjusted R^2	0.873
Residual Std. Error	4.312 (df = 507)
F Statistic	257.092*** (df = 14; 507)

Note: $^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$

Model comparison statistics also support the inclusion of the interaction term. A nested F-test shows that the interaction model fits significantly better than the corrected base model and the AIC further decreases from 3027.86 to 3023.88, indicating an improved explanatory performance.

Substantively, the findings suggest that while a greater instructional spending per pupil does not strongly correlate independently with student achievements on average, its effectiveness is conditioned by the extent of federal support. In other words, in states with more reliance on federal funding for education, increases in spending on instruction are more strongly associated with higher NAEP performance.

Taken together, the moderation analysis suggests that funding structure is an important ingredient in the relationship between education finance and student achievement, independent of spending levels alone. These findings motivate a broader discussion of how federal and state funding interact to influence educational outcomes.

Discussion

This study examines how differences in state education finance structures are associated with student academic achievement, with a particular focus on instructional spending per pupil and reliance on federal education revenue. Using a general linear modeling framework applied to state-year NAEP data, the analysis yields three interrelated findings that together underscore the importance of funding structure rather than spending levels alone.

To begin with, the negative correlation between federal revenue share and NAEP performance is noteworthy. Federally funded states tend to experience lower average student achievement than states that derive their education funding from other sources. However, it is crucial to note that reliance on federal funding should not be used as an argument that federal

funding is detrimental to performance. On the whole, the provision of federal education aid in the U.S. is purposely aimed at states and districts with a relatively high socioeconomic disadvantage and have a long-standing record of low public investment. Therefore, the observed negative association should be characterized as a symptom of structural disparities between states, with federal revenue share acting in place of a policy failure.

Secondly, the lack of a strong, clear association between instructional spending per pupil and student achievement arises when incorporating the effects of fixed year and influential observations into the model. Though instructional spending is highly correlated to NAEP scores in bivariate analysis, the relationship disappears in the multivariate statistical model. This result is in line with the idea that spending levels may be necessary but not sufficient for better results unless these funds are allocated to different uses, implemented properly, and are provided additional support in the form of complementary policies.

Third, this study's moderation analysis shows that the power of instructional spending varies when it is considered in relation to the funding environment in which is it situated. The positive and statistically significant interaction between federal revenue share and instructional spending shows that, in the states with the highest share of federal education funding, there is a stronger link between instructional spending and achievement. In other words, federal assistance can reinforce the state's capacity to achieve the intended result and may work through stabilizing the budget, supporting compartmentalized programs, or adding supplementary resources in settings of need.

By doing so, it may be seen that funding structure is not solely concerned with the funding size but the structural characteristics as well. This means that an evaluation that simply asks if higher spending leads to better outcomes will leave many results hidden. The federal

funding program seems to have a modified impact that affects how instructional investments are translated into achievement outcomes in an indirect manner.

Policy Implications

In that sense, the study encourages the policymakers to move beyond the aggregated spending levels and focus on more system questions of finance, like how the budget is utilized and how it is aimed towards the particular outcomes. Federal education funding can thus be most beneficial when it is intended to support, not replace, the funding for instructional investments carried out by other entities of state. The policy best practices do not include uniform increments of federal aid spending but align it with state instructional priorities to maximize their benefits, especially in the more needy contexts.

At the same time, the result of precluding a strong independent impact of instructional spending implies that it is a risk to assume that revenue increases will automatically lead to improved outcomes. The found effective use of schools' funds is likely to be influenced by factors, e.g., institutional capacity, quality of workforce, and program design, all of which have not been taken into consideration in this analysis.

Limitations

Some of the weaknesses in this approach need to be acknowledged. This study is an observational one, and it is not able to prove that finance is causing a change in achievement. State aggregation incorporates the all-local differences and inequality; also key covariates such as poverty rates, demographic mix, and specific policies are not available. Furthermore, only NAEP data can be used to analyze this period, which can produce anomalies in event timing.

Notwithstanding these problems, the study proves the efficacy of data-driven modeling process, and it emphasizes the relevance of understanding the role of spending in the larger context of funding structures.

Conclusion

This study examined how state-level education finance structures relate to student academic achievement, using a general linear modeling framework applied to state-year NAEP data from 1992 to 2015. The results indicate that while instructional spending per pupil is not independently associated with higher average achievement, its relationship with student performance depends on states' reliance on federal education funding. Federal revenue share is strongly associated with lower observed achievement levels, likely reflecting the targeted nature of federal aid toward higher-need states rather than a negative policy effect. Perhaps importantly, moderation analyses suggest that in these contexts, federal support may enhance the effectiveness of instructional spending. Together, these findings emphasize the importance of considering funding structure alongside funding levels in education policy analysis and warn against blanket conclusions about the role of school spending in efforts to improve student outcomes.

References

Baker, Bruce D. “Educational Inequality and School Finance: Why Money Matters for America’s Students.” *Harvard Education Press*, Harvard Education Press. 8 Story Street First Floor, Cambridge, MA 02138. Tel: 888-437-1437; Tel: 617-495-3432; Fax: 978-348-1233; e-mail: hepg@harvard.edu; Web site: http://hepg.org/hep-home/home, 30 Sept. 2018, eric.ed.gov/?id=ED594085.

Hanushek, E. A. (2003). The failure of input-based schooling policies. *The Economic Journal, 113*(485), F64–F98. https://doi.org/10.1111/1468-0297.00099

Jackson, C. Kirabo, et al. “The Effects of School Spending on Educational and Economic Outcomes: Evidence from School Finance Reforms.” *The Quarterly Journal of Economics*, President and Fellows of Harvard College, 1 Jan. 1970, econpapers.repec.org/article/oupqjecon/v_3a131_3ay_3a2016_3ai_3a1_3ap_3a157-218..htm.

Appendix 1: R code

```
df_raw <- readr::read_csv("/Users/leeeileen/Downloads/archive/states_all.csv
") %>%
  clean_names()
df <- df_raw %>%
  mutate(
    naep_composite = rowMeans(
      cbind(avg_math_4_score, avg_math_8_score, avg_reading_4_score, avg_read
ing_8_score),
      na.rm = TRUE
    ),
    # Per-pupil instructional spending
    instr_pp = instruction_expenditure / grades_all_g,
    # Log-transform spending to reduce skew
    log_instr_pp = log(instr_pp),
    # Federal revenue share
    fed_share = federal_revenue / total_revenue
  ) %>%
  filter(!is.na(naep_composite)) %>%
  filter(
    is.finite(log_instr_pp),
    is.finite(fed_share),
    instr_pp > 0,
    fed_share >= 0, fed_share <= 1
  ) %>%
  filter(year >= 1992, year <= 2015)
model_df <- df %>%
  select(state, year, naep_composite, log_instr_pp, fed_share) %>%
  drop_na()
nrow(model_df)
desc_tbl <- model_df %>%
  pivot_longer(cols = c(naep_composite, log_instr_pp, fed_share),
               names_to = "variable", values_to = "value") %>%
  group_by(variable) %>%
```

```
  summarise(
    N = sum(!is.na(value)),
    mean = mean(value, na.rm = TRUE),
    sd   = sd(value, na.rm = TRUE),
    min  = min(value, na.rm = TRUE),
    max  = max(value, na.rm = TRUE),
    .groups = "drop"
  )
desc_tbl
# outcome distribution
hist(model_df$naep_composite, breaks = 30, main = "NAEP composite", xlab = "S
core")
# spending per pupil distribution (raw vs log)
hist(df$instr_pp, breaks = 30, main = "Instructional $ per pupil (raw)", xlab
 = "instr_pp")
hist(model_df$log_instr_pp, breaks = 30, main = "log(instr_pp)", xlab = "log_
instr_pp")
# federal share
hist(model_df$fed_share, breaks = 30, main = "Federal revenue share", xlab =
"fed_share")
cor_mat <- cor(model_df %>% select(naep_composite, log_instr_pp, fed_share),
               use = "complete.obs")
round(cor_mat, 3)
m1 <- lm(naep_composite ~ log_instr_pp + fed_share + factor(year), data = mod
el_df)
summary(m1)
# Robust SE (HC1)
coeftest(m1, vcov. = vcovHC(m1, type = "HC1"))
p1 <- ggplot(model_df, aes(x = log_instr_pp, y = naep_composite)) +
  geom_point(alpha = 0.5) +
  theme_classic() +
  geom_smooth(method = "loess", se = TRUE) +
  geom_smooth(method = "lm", se = FALSE)
p1
p2 <- ggplot(model_df, aes(x = fed_share, y = naep_composite)) +
  geom_point(alpha = 0.5) +
```

```
  theme_classic() +
  geom_smooth(method = "loess", se = TRUE) +
  geom_smooth(method = "lm", se = FALSE)
p2
plot_df <- tibble(
  fitted = fitted(m1),
  resid  = resid(m1),
  abs_resid = abs(resid(m1))
)

ggplot(plot_df, aes(x = fitted, y = resid)) +
  geom_point(alpha = 0.5) +
  theme_classic() +
  geom_smooth(method = "loess", se = TRUE) +
  geom_smooth(method = "lm", se = FALSE)
ggplot(plot_df, aes(x = fitted, y = abs_resid)) +
  geom_point(alpha = 0.5) +
  theme_classic() +
  geom_smooth(method = "loess", se = TRUE) +
  geom_smooth(method = "lm", se = FALSE)
hist(resid(m1), breaks = 30, main = "Residuals", xlab = "Residual")
qqnorm(rstandard(m1)); qqline(rstandard(m1))
influence_measures <- tibble(
  state = model_df$state,
  year  = model_df$year,
  cooks_d = cooks.distance(m1),
  leverage = hatvalues(m1),
  stud_resid = rstudent(m1)
) %>%
  arrange(desc(cooks_d))
head(influence_measures, 10)
# Rule-of-thumb flags
n <- nrow(model_df)
flagged <- influence_measures %>%
```

```
  filter(cooks_d > (4 / n) | abs(stud_resid) > 3)
flagged
if (nrow(flagged) > 0) {
  model_df2 <- model_df %>%
    anti_join(flagged %>% select(state, year), by = c("state", "year"))
} else {
  model_df2 <- model_df
}
m1_corr <- lm(naep_composite ~ log_instr_pp + fed_share + factor(year), data
= model_df2)
AIC(m1, m1_corr)
summary(m1_corr)
m1_vif <- lm(naep_composite ~ log_instr_pp + fed_share, data = model_df2)
vif(m1_vif)
m2 <- lm(naep_composite ~ log_instr_pp * fed_share + factor(year), data = mod
el_df2)
summary(m2)
# Compare base vs interaction (nested F-test)
anova(m1_corr, m2)
# AIC comparison
AIC(m1_corr, m2)
modelsummary(
  list(
    "Base (Year FE)" = m1_corr,
    "Interaction (Year FE)" = m2
  ),
  statistic = "({std.error})",
  stars = TRUE
)
```